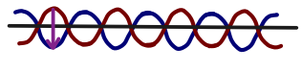
Dirac Equation: Relativity + Quantum Mechanics

In general, we want a Relativistic model of QM for fermions.

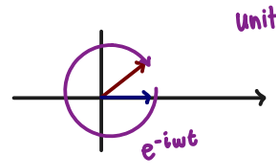
recall,



plane of light has $E(\vec{x},t)=0$ everywhere @ some t

→ Dirac wanted unitary dynamics linear derivative in time:

$E = \hbar\omega \rightarrow \frac{i\hbar\, d\psi}{dt} \rightarrow i\hbar\omega$



unitary evolution

Relativistic Invariant:

$E^2 = (mc^2)^2 + (pc)^2$

↳ $(E^2 + \Delta^2) = 0$

↳ $(E+\Delta)(E-\Delta) = 0$

Dirac: $E^2 = (mc^2)^2 + (pc)^2$

$$E^2 = (mc^2)^2 + (pc)^2 = 0$$

$$(E - \overleftrightarrow{\alpha}_0 mc^2 - c\overleftrightarrow{\alpha}\cdot\vec{p})(E + \overleftrightarrow{\alpha}_0 mc^2 + c\overleftrightarrow{\alpha}\cdot\vec{p})$$

– only works if $\overleftrightarrow{\alpha}$ matrices have the property: $\overleftrightarrow{\alpha}_i\overleftrightarrow{\alpha}_j = -\overleftrightarrow{\alpha}_j\overleftrightarrow{\alpha}_i$ if $i \neq j$

$\overleftrightarrow{\alpha}_i\overleftrightarrow{\alpha}_j = \alpha_i^2 = 1$

In general: α's do not commute: $AB \neq BA$

$\overleftrightarrow{\alpha}_i$ is a 4x4 matrix

$[\overleftrightarrow{\alpha}_0, \overleftrightarrow{\alpha}_x, \overleftrightarrow{\alpha}_y, \overleftrightarrow{\alpha}_z]$

(p_0, p_x, p_y, p_z)

$\overleftrightarrow{\alpha}\cdot p = \overleftrightarrow{\alpha}_x p_x + \overleftrightarrow{\alpha}_y p_y + \overleftrightarrow{\alpha}_z p_z$

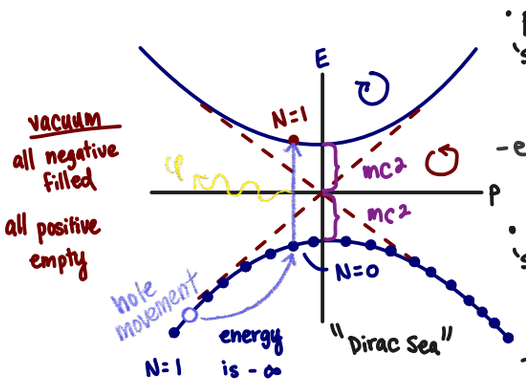
- $\overleftrightarrow{\alpha}_0 = \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & \\ \hline 0 & & \begin{matrix} -1 & 0 \\ 0 & -1 \end{matrix} \end{array}\right)$

- $\overleftrightarrow{\alpha}_i = \left(\begin{array}{c|c} 0 & \overleftrightarrow{\sigma}_i \\ \hline \overleftrightarrow{\sigma}_i & 0 \end{array}\right)$

$\overleftrightarrow{\sigma}_x = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\overleftrightarrow{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$\overleftrightarrow{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$



- positive energy 'spin up or down'

Pauli Exclusion
– each state can have only N=0 or N=1

- negative energy 'spin up or down'

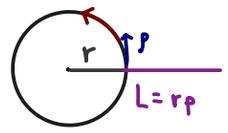
↳ spin statistics thm: two values of N=0 or 1 connected to 2 choices of spin in $\psi(x)$

"hole" – acts as a particle with positive charge + positive mass – antimatter positron

introduces new degree of freedom some index which the relativistic Schrödinger eqn relies on:

"spin"

What is spin? 'angular momentum in QM'



$\vec{L} = \vec{r}\times\vec{p}$

QM: $\vec{p} = -i\hbar\vec{\nabla}$

$L = rp$

boundary condition:

$\psi(x) = \psi(x + 2\pi r)$

$\rightarrow \lambda = \frac{2\pi r}{N}$ $(2\pi r = N\lambda)$

$L = r(\hbar k)$
$= r\hbar\left(\frac{2\pi}{2\pi r}\right)N$
$= \hbar N$ ← units of angular momentum

'orbital angular momentum' $k = \frac{2\pi}{\lambda} = \frac{2\pi}{2\pi r}N$

⇒ worked example: $[L_x, L_y] = [yp_z - zp_y, zp_x - yp_z]$

$$\left.\begin{aligned} &= yp_z zp_y - yp_z xp_z \\ &- zp_y zp_x + zp_y xp_z \\ &- zp_x yp_z + zp_x zp_y \\ &- xp_z yp_z - xp_z zp_y \end{aligned}\right\} = (p_z z - zp_z)(yp_z - xp_y)$$

$p_z z - zp_z = -i\hbar$, $yp_z - xp_y = L_z$

↳ $[L_x, L_y] = i\hbar L_z$

Connection of spin to angular momentum

$\vec{L} = \vec{r}\times\vec{p}$

in Q.M.: $-i\hbar\vec{\nabla}$ ← an operator

recall: many operators do NOT commute $(AB = BA)$

$[A, B] \equiv AB - BA$

* e.g. Commutator of x, p

$-1 = [x, \frac{d}{dx}]f(x)$

↳ $= (x\frac{d}{dx} - \frac{d}{dx}x)f(x)$

$= x\frac{df}{dx} - [f(x) + x\frac{df}{dx}] = -f(x)$

Pauli Matrices for Spin

$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

define spin matrix:

$S_x = \frac{-\hbar}{2}\sigma_x$, $S_y = \frac{\hbar}{2}\sigma_y$, $S_z = \frac{\hbar}{2}\sigma_z$

↳ $[S_x, S_y] = \frac{\hbar^2}{2y}2i\sigma_z$

$= i\hbar(\frac{\hbar}{2}\sigma_z)$

$= i\hbar S_z$

$L = \hbar N$

$\sigma_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

eigenstates are: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

eigenenergies are: $\frac{\hbar}{2}, \frac{-\hbar}{2}$: $S = \frac{\hbar}{2}$

$[\sigma_x, \sigma_y]$:

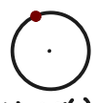
$$\sigma_x\sigma_y - \sigma_y\sigma_x \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = 2i\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = 2i\sigma_z$$

L

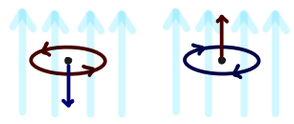
$\psi(x)$

⇒

spin $\frac{\hbar}{2}(\pm 1)$



$\psi(x)e^{i\theta(x)}$



define: anticommutator

↳ $\{A, B\} = AB + BA$

suppose $\{A, B\} = 0$

$AB = -BA$

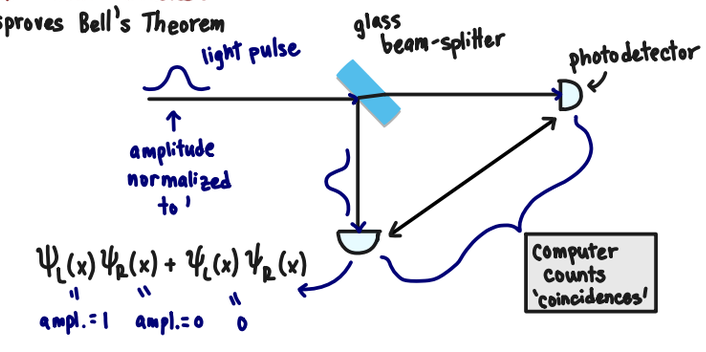
operations: ① measure particle 1
② ↓ 2

2 full rotations for particle to return to starting position!?

fermions (matter particles): spin $\frac{1}{2}$

Copenhagen "Knowledge leads to collapse"

example of entangled state

Local Hidden Variables:

→ disproves Bell's Theorem